

IAN MERRIMAN

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EDUCATION

University of Texas at Arlington

Arlington, TX

Bachelor of Science, Aerospace Engineering | GPA: 3.6 | Expected Graduation: August 2026

Relevant Coursework: Astronautics I & II, Spacecraft Dynamics and Control, Space Systems Engineering, Space Environment, Automatic Controls, Flight Dynamics, Flight Performance, Heat Transfer, Mechatronics

TECHNICAL SKILLS

Programming & Software: MATLAB/Simulink, Python/MicroPython, C++, Git/GitHub

Aerospace Analysis: Trajectory Modeling, 6-DOF Flight Dynamics, Stability Analysis, Drag Modeling

Simulation & Controls: PID Control, Control System Design, State-Space Modeling, Simulink Modeling

Design & Manufacturing: SolidWorks, Recovery System Design, Composite Layups, Soldering

RELEVANT PROJECTS

Thrust-Vectoring Rocket Project – Flight Controls Team

July 2025 – Present

- Developed a MATLAB/Simulink PID control model for a thrust-vectoring rocket landing system, using body angle and angular-rate feedback to command gimbal angle corrections.
- Designed controller gains for an estimated critically damped attitude response, enabling simulation-based evaluation of landing stability before vehicle construction.
- Tested controller response across multiple initial body angles and angular rates to evaluate closed-loop behavior under varying landing conditions.

High-Power Rocketry Projects – Aero Mavs

Sept 2022 – July 2024

- Designed, built, and launched two H-class personal rockets using OpenRocket and SolidWorks, achieving altitudes up to 1,850 ft with a calculated stability margin of 1.97 calibers.
- Designed and manufactured a 9.7 oz mechanical drogue-deployment nose cone using motor-burnout ejection to resolve previous recovery failures.
- Manufactured competition rocket components through epoxy layups, sanding, drilling, and fin fillet fabrication for two Aero Mavs launch vehicles.
- Contributed manufacturing work to competition rockets that achieved 12,200 ft at Mach 1.12 and 23,000 ft at Mach 1.7.

Self-Balancing Table Project – Controls Programmer

Jan 2025 – May 2025

- Programmed a two-axis PID control system in MicroPython on a Raspberry Pi Pico to stabilize a ball on a touchscreen using two servo-actuated tilting axes.
- Integrated touchscreen position feedback with real-time servo commands to maintain ball stability during tested operating conditions.
- Delivered a successful demonstration that met all tested design requirements and earned an A.

LEADERSHIP & ORGANIZATIONS

Aero Mavs – UTA Aerospace Student Organization

President

July 2024 – July 2025

- Oversaw operations, managed cross-disciplinary technical divisions, coordinated projects and competitions.
- Secured \$4,000+ in funding for competition expenses and doubled active membership from 53 to 110.

AIAA at UTA – Project BlueSpear

Navigation Lead

Feb 2026 – Present

- Serve as the primary trajectory simulation developer for a student-led 100 km Kármán-line rocket project.
- Developed a 13-state MATLAB 6-DOF trajectory simulation for Project BlueSpear, propagating vehicle position, velocity, quaternion attitude, and body angular rates.
- Mentored freshman and sophomore team members in flight dynamics to support trajectory modeling efforts.
- Led team through 5 major simulation iterations from 3-DOF modeling to thrust/mass integration, atmospheric modeling, 6-DOF dynamics, and visualization tools.